

CLASS 02 / SEP 1

TRANSFORMATIONS AND CSG

P1 BEGINS — DESCRIBE A SOLID AS AN EDITABLE SEQUENCE

NOTHING DUE

P1a ISSUED TODAY

WHY NON-DESTRUCTIVE

The same two solids, subtracted. One keeps its history; one throws it away.

DESTRUCTIVE

CUBE - SPHERE → ONE MESH

The sphere is gone. Its radius is gone. To change anything you rebuild from scratch.

NON-DESTRUCTIVE

CUBE + SPHERE → SUBTRACT → RESULT

Every input is still a live node. Change the radius and everything downstream recomputes itself.

P1a REQUIRES YOUR OPERATION HISTORY TO BE INTACT. YOU WILL BE ASKED TO CHANGE A PARAMETER IN FRONT OF THE CLASS.

TRANSFORMATIONS ARE ARITHMETIC

Three operations you already know, written as what they actually do.

MOVE

= ADD

$$x' = x + dx$$

$$y' = y + dy$$

$$z' = z + dz$$

$$(2, 1) + (3, 0) \rightarrow (5, 1)$$

SCALE

= MULTIPLY

$$x' = x \times f$$

$$y' = y \times f$$

$$z' = z \times f$$

$$(2, 1) \times 2 \rightarrow (4, 2)$$

ROTATE

= MIX x AND y

$$x' = x \cdot \cos\theta - y \cdot \sin\theta$$

$$y' = x \cdot \sin\theta + y \cdot \cos\theta$$

$$(1, 0) \text{ rot } 90^\circ \rightarrow (0, 1)$$

Move and scale treat each coordinate on its own. Rotation is the only one where x and y have to talk to each other.

THREE BOOLEANS

Everything in P1a is built from these, applied in order.

UNION

$$A + B$$

Everything either solid occupies.

DIFFERENCE

$$A - B$$

A, with B carved out of it.

INTERSECTION

$$A \cap B$$

Only where both solids overlap.

ORDER MATTERS. $A - B$ IS NOT $B - A$, AND YOUR DIAGRAM MUST MAKE THE OPERANDS UNAMBIGUOUS.

P1a asks for at least three Booleans, so one of them can be a reversed subtraction. Do it once deliberately and you will never confuse the operands again.

THE SEQUENCE

What we build together, in order, this session.

01

Introduce the node editor, inputs, outputs and parameters using one primitive.

02

Move, rotate and scale procedurally; change the order and compare the result.

03

Compare union, difference and intersection between two solids, including reversing the subtraction.

04

Expose useful inputs and compose a short massing sequence.

THE NOTATION

Write the procedure as pseudocode. Formal enough to be unambiguous, plain enough to read aloud.

THE OPERATIONS

```
box(x, y, z)      cylinder(r, h)
sphere(r)         cone(r, h)

move(s, x, y, z)  rotate(s, axis, deg)
scale(s, factor)

union(a, b)       subtract(a, b)
intersect(a, b)
```

WORKED EXAMPLE

```
base  = box(6, 4, 3)
tower = box(2, 2, 6)
tower = move(tower, 2, 0, 3)
void  = cylinder(r=1, h=5)
void  = move(void, -1.5, 0, 0)

mass  = union(base, tower)
result = subtract(mass, void)
```

THREE RULES: EVERY SHAPE IS BORN AT THE ORIGIN · UNITS ARE ABSTRACT · EVERY STEP ASSIGNS A NAME, SO A SUBTRACTION IS NEVER IN DOUBT.

THE SAME THING, WRITTEN OUT

You are not asked to type this. Read it, and check that it says what the nodes you just built say.

```
# nine diagrams, one operation each, in this order
```

```
box      = cube(1.0)           # 1  primitive
ball     = sphere(r=0.65)      # 2  primitive
rod      = cylinder(r=0.30, h=2.0) # 3  primitive

ball     = scale(ball, 1.15)    # 4  scale
ball     = translate(ball, z=0.15) # 5  translate
rod_y    = rotate(rod, x=90)    # 6  rotate - degrees
rod_x    = rotate(rod, y=90)    # 6  same operation

solid    = intersection(box, ball) # 7  intersection
cutters  = union(rod, rod_y, rod_x) # 8  union
result   = difference(solid, cutters) # 9  difference
```

```
# 9 keeps solid, removes cutters. draw 9 again at
# building scale: a figure, planting, ground, shadow.
```

WHAT IT MAPS ONTO

parameter
vector
translation
rotation
scale
union
difference
intersection

THIS IS THE NOTATION FROM THE LAST SLIDE, AND IT IS ALSO PYTHON. MORE ON THAT IN DECEMBER.

ISSUED TODAY

P1a — CSG MASSING SEQUENCE

DUE WEEK 4 — Sep 15 / 7%

Build a parametric massing study, then build a repeating panel from its own points and faces and develop a small printable variant of it. This first milestone is the massing study: an ordered sequence of primitives and Booleans, diagrammed so another person could follow it.

STAGE 1

THE SEQUENCE

Three primitives, three transformations, three Booleans, in a stated order.

STAGE 2

THE DIAGRAMS

Eight or nine, numbered. One operation each, read left to right.

STAGE 3

THE LAST ONE

Drawn, not diagrammed: building scale, a figure, planting, ground, shadow.

REQUIREMENTS — P1a

Visual quality 35% · computational understanding 25% · technical execution 20% · experimentation 10% · requirements and identity 10%.

01

One 17 × 11 inch sheet carrying an ordered sequence of eight or nine numbered diagrams: one operation per diagram, read in order, each showing what the one before it produced.

02

Use at least three primitives, at least three transformations covering translate, rotate and scale, and at least three Boolean operations covering union, difference and intersection. Name the operands of every subtraction — which solid is kept and which is removed.

03

Develop the last diagram as a drawing rather than a diagram: the final solid at building scale, with a 1.75 m human figure, vegetation, ground and shadow, so the sequence ends in something readable as architecture.

04

Write the sequence out as pseudocode beside the diagrams — one line per operation, in the same order as the diagrams, with the parameters you exposed.

05

Read the spatial operations in Di Mari and Yoo, *Operative Design*, and Di Mari, *Conditional Design*, and state which operation your sequence performs in their terms.

06

Submit an editable graph with exposed inputs and geometry generated in the graph. Keep the source operations intact.

NOW — BUILD THE SEQUENCE

Rest of the session: a working graph with something you can change in front of me.

✓

Object chosen — orthogonal massing, 5–10 clear moves, no complex curves

✓

First three primitives placed, each born at the origin and moved from there

✓

At least one Boolean working, with the operands named

✓

Two useful inputs exposed, so a variant is one number away

✓

Operation history intact — nothing collapsed or applied

LEAVE WITH A GRAPH YOU CAN CHANGE, NOT A MESH YOU CANNOT.

BEFORE NEXT CLASS

WATCH

U03 — Geometry from scratch. Point, line, edge, face, mesh, solid. The Blender lesson file for this one already exists; check the unit page.

BUILD

Keep going on P1a. It is due Week 4 — eight or nine numbered diagrams, the last one developed in context, and the sequence written out as pseudocode.

WRITE

Your procedure as pseudocode, using the notation above. Read it aloud to someone who has not seen your model and fix what they cannot follow.

BRING

The editable file, and one question about something in it that behaves oddly.

READ

Di Mari and Yoo, Operative Design, and Di Mari, Conditional Design. Find the spatial verb your own sequence performs, and name it on the sheet.